UNIVERSITY of WASHINGTON

# Software Design for Data Science

## Advanced Packages

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# Class Updates

Next week: Practice Presentations

- Similar to your first demo - low stress
- You should ALSO have a first draft of your presentation
  - Definitely does not need to be complete, but we want to see the beginning
- This is also your time to have any last minute discussions with instructors

# Project Requirements: stuff we've learned

Just to reiterate: you must include the following in your projects.

- **Tests**
  - High test coverage
  - High coverage is not *sufficient* - make sure you write good tests
- **README**
  - Updated!
- **Inline documentation (docstrings)**
  - For functions/modules
- **examples** folder
  - How do various users interact with your system?
  - How do we (the instructors) run the code and tests?
- **Style**: pylint 100%
- **Continuous Integration**
- **Pull requests & reviews**
- **Packaging** *(today!)*

# Distributing a Python application

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pandas-dev / pandas Public

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main | pandas / README.md | Go to file

MarcoGorelli CI remove scorecards (#50269) ... ✕ Latest commit f759c33 on Dec 15, 2022 History

57 contributors +39

170 lines (132 sloc) | 10.1 KB | Raw | Blame

pandas

# pandas: powerful Python data analysis toolkit

# PyPI

pandas · PyPI

https://pypi.org/project/pandas/

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## pandas 1.5.3

```
pip install pandas
```

Latest version

Released: Jan 19, 2023

Powerful data structures for data analysis, time series, and statistics

**Navigation**

- Project description
- Release history
- Download files

**Project links**

- Homepage
- Bug Tracker
- Documentation
- Source Code

**Project description**

pandas

pandas: powerful Python data analysis toolkit

# Extending project structure for PyPI

```
myproject/
    README.md
    LICENSE
    pyproject.toml
    myproject/
        __init__.py
        core.py
        subpackage/
        submodule/
        tests/
```

Contains metadata for the package for PyPI. This metadata is then piped to a package builder - often `setuptools` but there are others like `hatchling`

https://packaging.python.org/en/latest/tutorials/packaging-projects/

# Extending project structure for PyPI

```
myproject/
    README.md
    LICENSE
    pyproject.toml
    requirements.txt
    myproject/
        __init__.py
        core.py
        subpackage/
        submodule/
        tests/
```

Contains absolute dependencies, especially useful if you're publishing an application.
Can be generated using:

```
conda list --export > requirements.txt
```

OR

```
pip freeze > requirements.txt
```

# Extending project structure for PyPI

```
myproject/
    README.md
    LICENSE
    pyproject.toml
    requirements.txt
    MANIFEST.in
    myproject/
        __init__.py
        core.py
        subpackage/
        submodule/
        tests/
```

Specify data and files that should also be packaged in addition to the Python modules

# Example `pyproject.toml`

https://github.com/UWDATA515/ci_example/blob/main/pyproject.toml

# More examples

Check out existing Python packages. These may be more complicated:

https://github.com/numpy/numpy

https://github.com/pandas-dev/pandas

https://github.com/pylint-dev/pylint

# Submitting your package to PyPI

Update your code and version number. Run your test suites and ensure your code works as intended. Create a PyPI account if you don't have one already

Create your source, and if desired, binary distribution:

```
$ python -m build
```

Install `twine` package to submit builds to PyPI.

(Can install using conda install twine, pip install twine, etc.)

```
$ twine upload dist/*
```

**NOT REQUIRED!**

# Exercise: Set up your project's `pyproject.toml` file

https://setuptools.pypa.io/en/latest/userguide/index.html

Make sure it builds!

```
python -m build
```

(remember to install the build tools as part of your environment!)

Test it out!

- Move the built tar.gz from dist to a folder outside your repo
- Create a new (empty) conda environment
- `python3 -m pip install ./<your-package-name>.tar.gz`
- Try running python and try importing your modules and functions